

Jerry Xu

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PROFESSIONAL SUMMARY

Senior Software Engineer specializing in large-scale distributed systems and control plane architecture. Leads high-impact infrastructure initiatives on Azure's SDN platform, improving reliability, scalability, and performance across millions of resources. Proven ability to drive cross-organizational designs and deliver production-critical systems at global scale.

PROFESSIONAL EXPERIENCE

Microsoft | Azure SDN Control Plane

2019 – Present

Software Engineer → Software Engineer II (2020) → Senior Software Engineer (2022)

- Drove the integration of regional network management into Azure's new SDN control plane, implementing pub/sub-based goal-state publishing for VM, NIC, and network policy resources. Part of a multi-org effort that reduced VM lifecycle latency by 90%.
- Led cross-organizational design with Azure Compute to redesign the VM deletion API, eliminating resource-conflict incidents caused by premature SDN release and enabling migration of the VM lifecycle to a modern IaaS architecture.
- Redesigned goal-state distribution to route publish calls to VM home availability zones, reducing cross-AZ traffic by 66–75% and improving resilience during zone-level pubsub outages.
- Built a proactive error-triage workflow to surface low-frequency failures below alerting thresholds, evolving it from manual review to AI-driven classification and uncovering previously undetected production issues.
- Onboarded a new 6-person engineering team in India through structured weekly sessions and hands-on support, enabling independent contributions within two months. The team has since grown to 15+ engineers.

Capital One Canada | Software Developer

2018

- Designed and prototyped real-time event-driven data streams using Apache Kafka for financial ledger updates, architected for zero data loss across ~2M Canadian cardholders.

IBM | Software Developer

2017 – 2018

- Built and maintained robust backend services in Java/C++, focusing on enterprise-scale data processing, storage reliability, and security practices including OWASP principles and GDPR compliance.

EDUCATION

McMaster University | Bachelor of Science in Computer Science, Minor in Business

2017

Dean's Honor List

TECHNICAL SKILLS

C#, Java, Python, Azure, AWS, Apache Kafka, Git, GitHub, SDN, distributed systems, microservices, REST APIs